

Problem 5

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Problem 1 A table of values for f and f' is shown below. Suppose that f is a one-to-one function and f^{-1} is its inverse.

x	$f(x)$	$f'(x)$
1	3	4
3	4	5
4	6	3

- (a) Evaluate $f^{-1}(f(x))$ at $x = 3$.
- (b) Evaluate $\frac{d}{dx} f(f(x))$ at $x = 3$.
- (c) Evaluate $\frac{d}{dx} \ln(f(x))$ at $x = 3$. [2]
- (d) Evaluate $f^{-1}(x)$ at $x = 3$. [2]
- (e) Evaluate $\frac{d}{dx} f^{-1}(x)$ at $x = 3$. [2]
- (f) Evaluate $\lim_{x \rightarrow 4} \frac{f(x) - f(4)}{x - 4}$ [2]